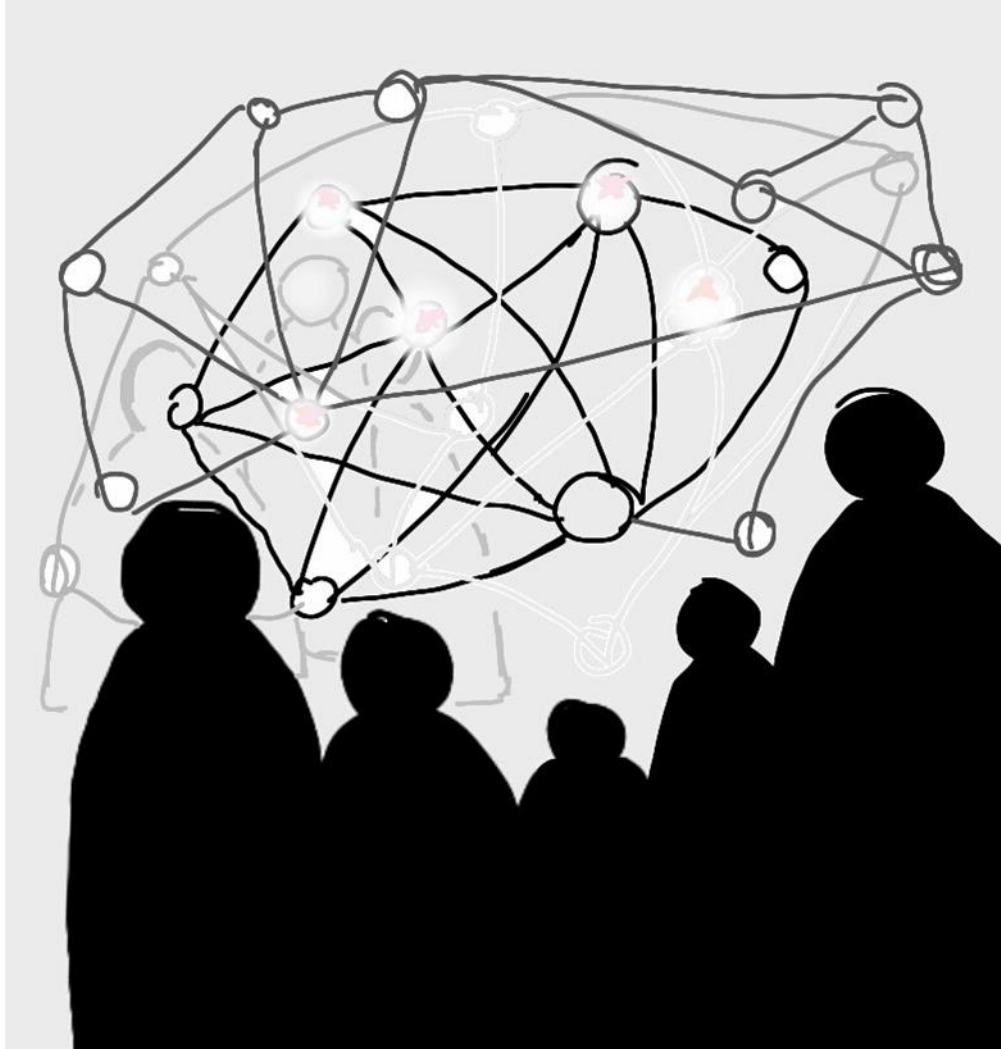


AI 507: Artificial Intelligence and Society



#12 AI in the workforce

Flashlight



Flashlight

AI in public administration

European principles

Certification

Factors motivating adaption

Odense in 2030

AI in political communication

What is political communication?

AI shapes political communication

GenAI in the Danish election

Danes attitudes towards GenAI in political communication

Mis- and Disinformation

Definition (and challenges)

Typical disinformation campaigns

Prevalence and effects

Two-factor models of persuasion

Elaboration-likelihood model

Systematic-Heuristic model

Susceptibility to misinformation

Lack of cognitive elaboration/
System I thinking

Motivated cognition

Conspiracy mentality

Red teaming AI against Denmark

**Do you have questions to
last week?**



Then let's get started!

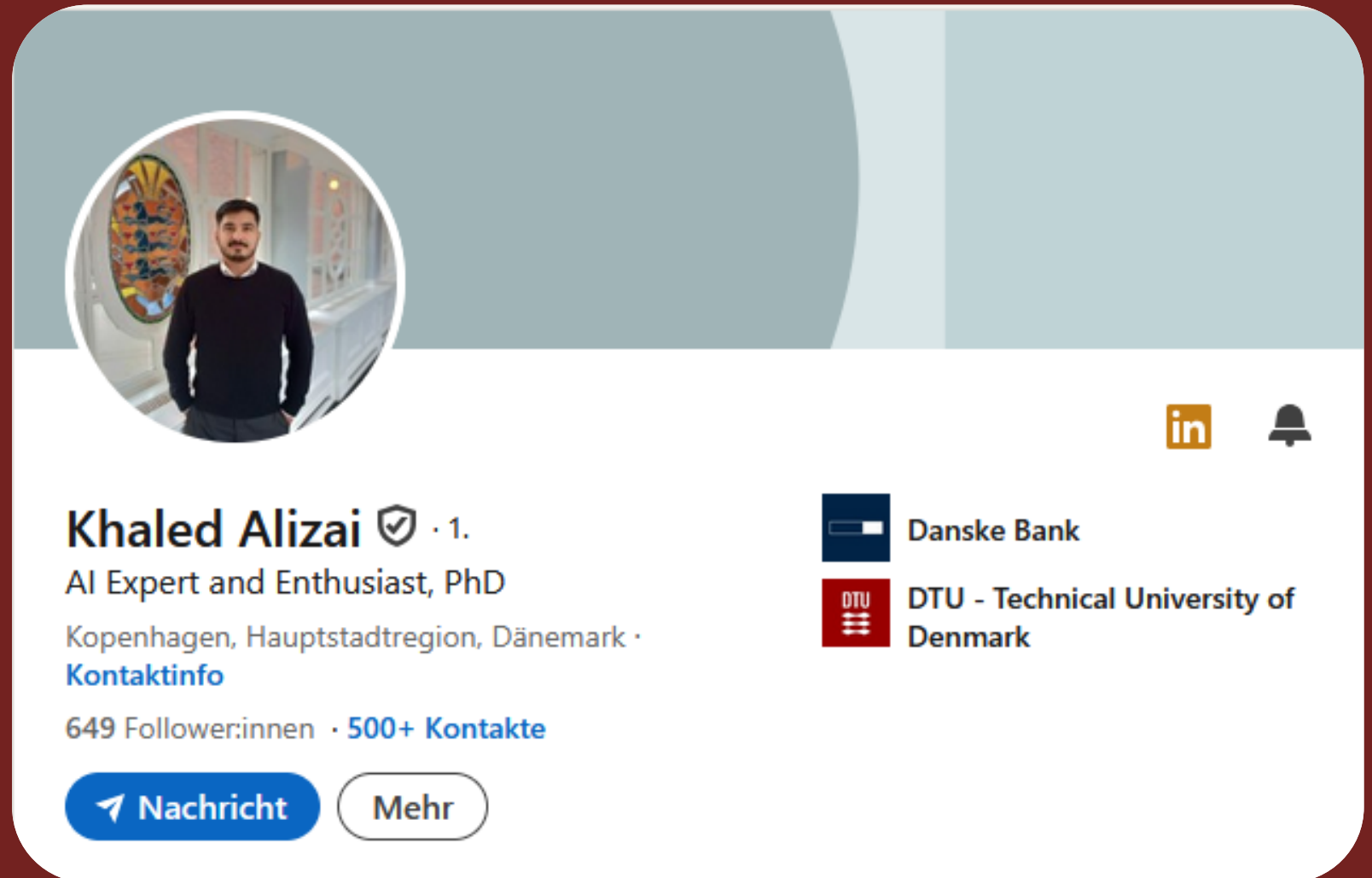
After today's lecture, you will hopefully have...

- ❑ ...learned how responsible AI in organisations can look like
- ❑ ...examine the specificities of the finance sector
- ❑ ... have first hand insights into institutions adapt to AI

To this end:

- Before the break: Guest talk from the Danske Bank
- After the break: Focus on AI in organisations and how AI is changing the workforce

Guest talk



A LinkedIn profile card for Khaled Alizai. The card features a circular profile picture of a man with a beard and a dark sweater. To the right of the profile picture are icons for LinkedIn and a notification bell. Below the profile picture, the name 'Khaled Alizai' is displayed with a verified badge and a '1' next to it. Underneath the name is the title 'AI Expert and Enthusiast, PhD' and the location 'Kopenhagen, Hauptstadtregion, Dänemark'. A link 'Kontaktinfo' is provided. Below this, it shows '649 Follower:innen' and '500+ Kontakte'. At the bottom, there are two buttons: 'Nachricht' (Message) and 'Mehr' (More).



Khaled Alizai  · 1.

AI Expert and Enthusiast, PhD

Kopenhagen, Hauptstadtregion, Dänemark · [Kontaktinfo](#)

649 Follower:innen · 500+ Kontakte

[Nachricht](#) [Mehr](#)



A blurred version of the LinkedIn profile card for Khaled Alizai, showing the same elements as the one above but with reduced contrast and detail.

[Nachricht](#) [Mehr](#)

649 Follower:innen · 200+ Kontakte

Time for a break



Artificial Intelligence at Work

Work?

(Deranty & Corbin, 2024)

- Quasi philosophical question
- Here: activities that individuals engage in as part of the production of goods and services – for a profit (business owners) or a wage (employees)
- Can be in the commercial, public, or “third”-sector (voluntary, NGO-work)
- Can also cover other activities and create other values (e.g., care work)
- But today we focus on traditional, restricted economic activities
- What role does AI have for the future of work?

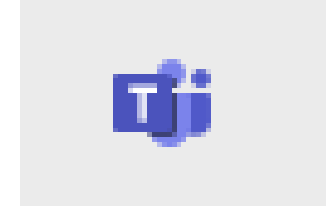


<https://easy-peasy.ai/ai-image-generator/images/ai-robot-revolutionizing-workplaces-diversity-reactions>

AI is not developed in a vacuum (Deranty & Corbin, 2024)

- AI innovation is (often) driven by attempts to win economic competition
- Sold using classic capitalist visions of innovation such as flexibility and adaptability
- Mundane mechanisms behind: Attempt to reduce costs, notably for labour and transactions,
- Increase outputs,
- Manage more efficiently through increased surveillance and control
- Create monopolies
- Plus: geopolitical tensions

What your employer can see on teams



public



private



deleted



Created content

public

private



Lena - Syddansk Universitet					
Home	Sort	View	...		
Name	Status	Date modified	Type	Size	
Apps		18.04.2024 15:23	File folder		
Microsoft Copilot-chatter		06.01.2025 08:55	File folder		
Microsoft Edge Downloads		23.07.2024 11:12	File folder		
Microsoft Teams Chat Files		21.01.2025 17:12	File folder		
Microsoft Teams-Chatstation		25.01.2024 09:27	File folder		
Microsoft Teams-chatfiles		12.05.2024 17:58	File folder		
Optagelser		17.03.2025 08:47	File folder		
Scans		14.01.2025 11:09	File folder		
Webbsteder filer		25.01.2024 09:27	File folder		
Whiteboards		25.01.2024 09:27	File folder		

The fact that what is called the technological development of modern times has been so largely oriented economically to profit-making is one of the fundamental facts of the history of technology

(Surveillance capitalism; *Zuboff*, 2019)



Shoshana Zuboff, via
https://de.wikipedia.org/wiki/Shoshana_Zuboff

What do we know about AI and unemployment?

(Deranty & Corbin, 2024; Klenert et al., 2023)

- Successful AI implementation = successful results of (former) human labour
- Prior innovations primarily affected unskilled labour
- That is not necessarily the case for AI
 - No evidence for robots reducing the share of low-skilled workers across Europe if the trend for decline is controlled for, new jobs emerge
 - For AI: non-linear relationship between AI patents and jobs: AI initially increases unemployment rates but – when the inflation is “normal” - unemployment shrinks
 - Central: that does not mean that people do not lose jobs, and the model does not account for crises and other “non-standard” times

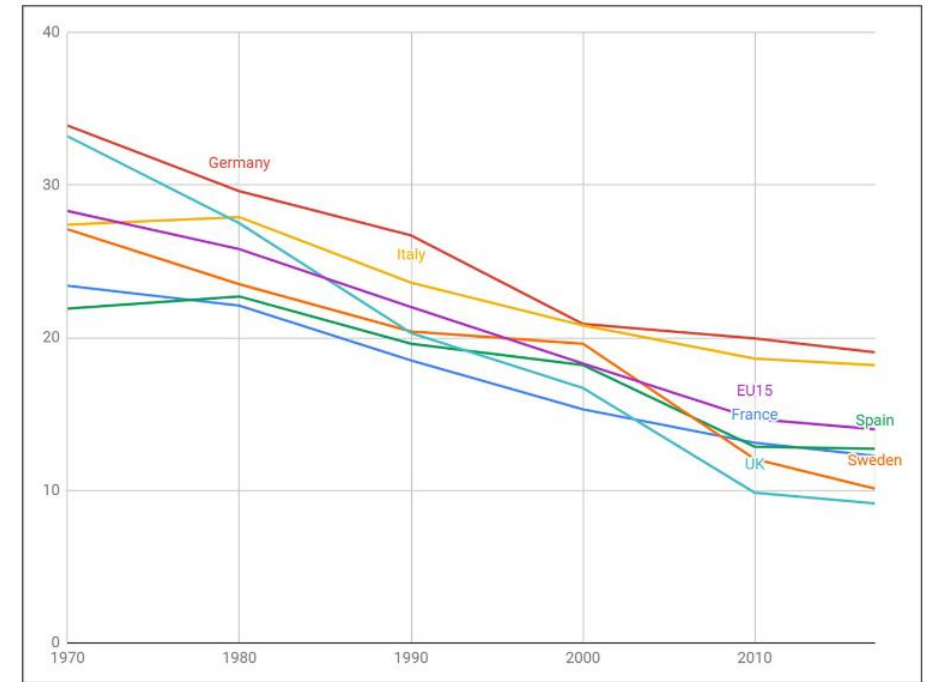


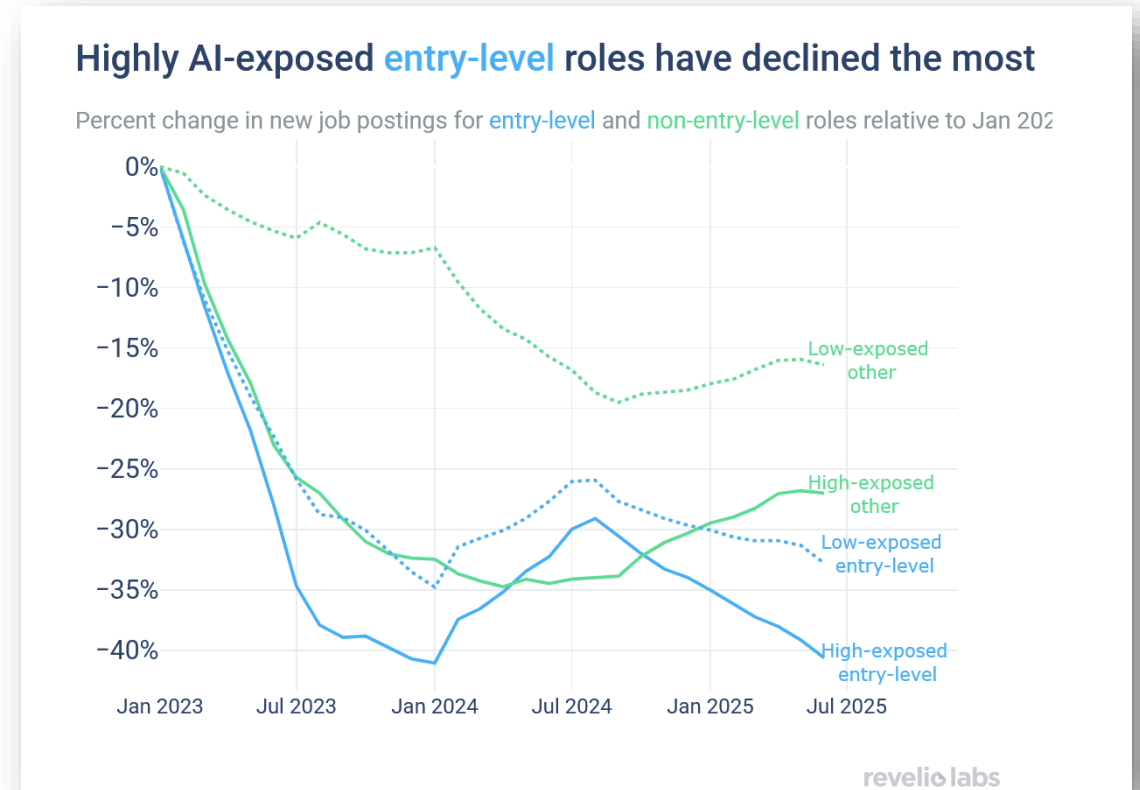
Figure 2. Share of employment in manufacturing in Europe (1970–2016).
Source: Authors' analysis from EU KLEMS database and EUROSTAT.

Klenert et al., 2023, S. 289

What do we know about how AI changes the workforce?

(Simon, 2025)

- AI affects specifically entry-level job opportunities
- Following an analysis of Revelio Labs and Bloomberg, sectors that have more tasks that can be done by AI have seen the largest decline
- Losses are unequally distributed:
 - Affects mostly entry-level jobs (expectation that seniors will adapt)
 - Does affect fields with higher AI potential more



What do we know about how AI changes the workforce? (Simon, 2025)

Higher AI exposure correlates with lower demand for entry-l

YoY change in job postings, by AI exposure rates for entry-level and non-entry-level jobs



- Coordinator: +17% AI exposure
- 75% more entry-level jobs

- Digital marketing: +25% AI exposure
- -5% less entry level jobs

- + 56% AI exposure for software developers
- - 67% of job-postings on the entry level

Those who are still needed are able to monitor and evaluate AI – and to translate between AI and humans

Which jobs are there around AI?

- Analysis of aggregated online job postings in the US 2021-2022
- Increased demand for AI jobs beyond computer science and being equipped with statistical competencies
- E.g., ~40% more jobs in AI + Statistics between 2021 and 2022 – particularly for
 - Data analysts
 - Machine learning engineers & scientists
 - Risk and business analysts
 - Marketing managers
 - Sustainable consultant

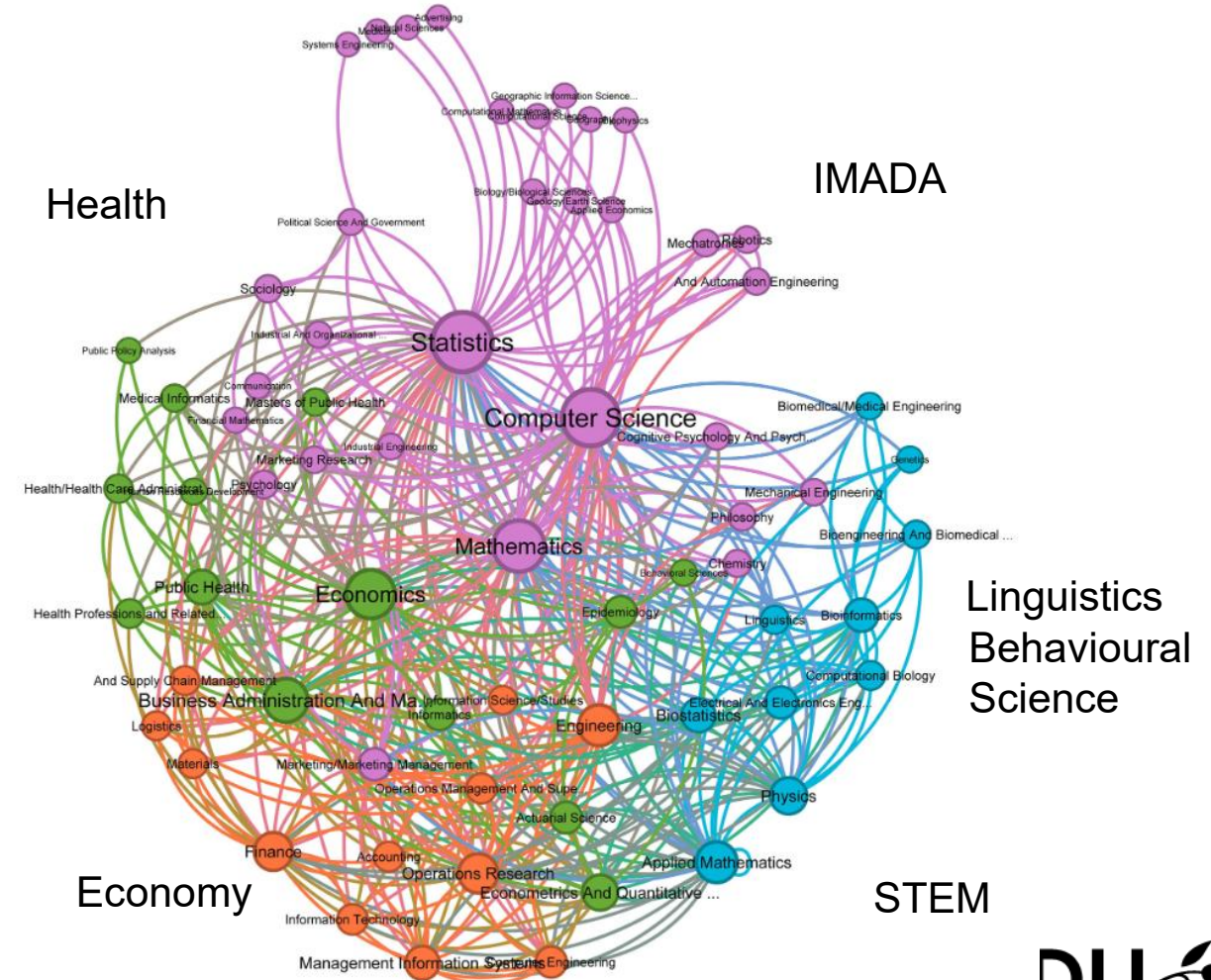
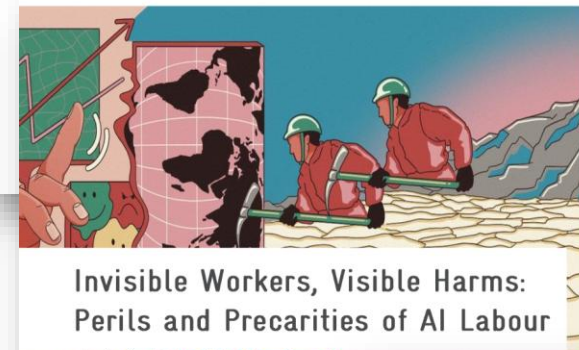
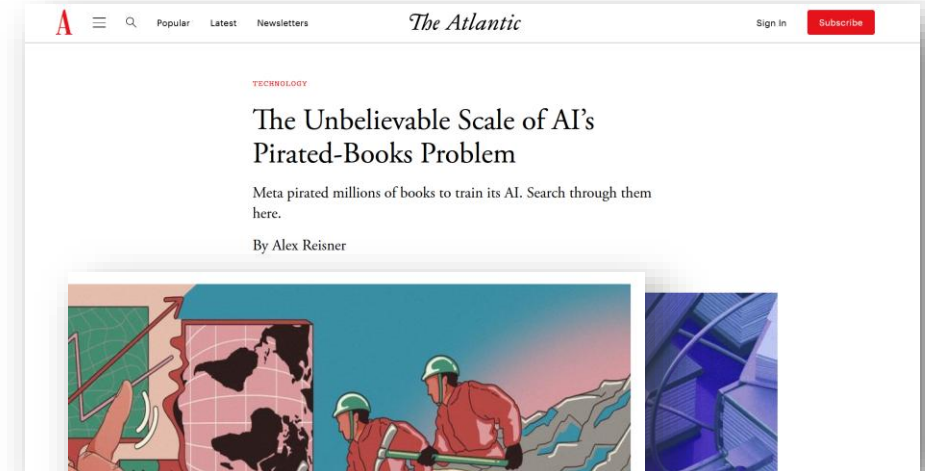


Fig. 4 Discipline cluster networks for statistics in AI recruitment.

Liu et al., 2024, S. 7

And of course: the hidden workforce

- High-quality “human-in-the-loop” training data requires humans
- Even modern deep-learning models do not train themselves (error-free)
- Three central “new” job markets:
 - Extraction of value by content creators (artists, scientists, journalists etc.) who are not paid at all
 - Low-paid crowd-workers often in the Global South –young, highly educated doing boring, traumatizing and incredibly low-paid crowd coding jobs (often ~1 -2 \$/ hour)
 - Highly skilled experts that fine-tune special models & management



Meet Mercy and Anita - the African workers driving the AI revolution, for just over a dollar an hour



© 2024, Jonathan 'Quentin' Gargano and 15 others. It is a large business process outsourcing centre. Photograph: Yusef Tilly/Getty Images

Social media content and AI training data are processed in outsource centres in the global south, where long hours, low pay and exposure to disturbing material are the norm

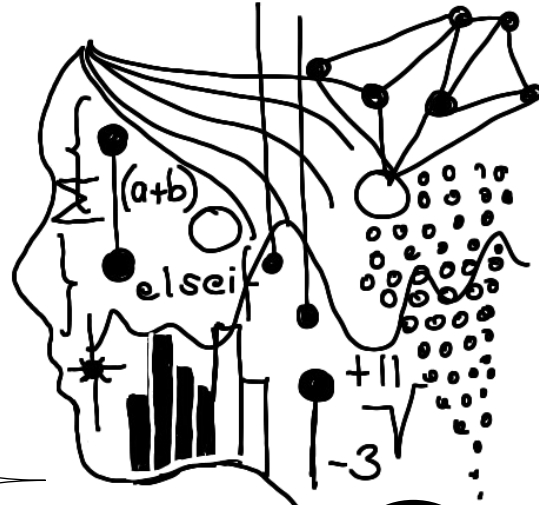
James Muldoon, Mark Graham and Callum Cant: 'AI feeds off the work of human beings'

AI and management

(Deranty & Corbin, 2024; Mateescu & Nguyen, 2019)

Typically tasks of a manager

- Hiring
- Optimisation of the work process
- Evaluation of workers
- Scheduling of shifts
- Coordinating customer demands with service providers
- Monitoring work
- Incentivisation



What could go wrong?

Project Management



What my friends think I do.



What my mom thinks I do.



What society thinks I do.



What my boss thinks I do.



What I think I do.



What I actually do.

<https://de.pinterest.com/pin/539869074050893004/>

- Surveillance and control
- Power imbalances
- Bias and discrimination
- Accountability of the management

What skills do leaders need now?

(Bock & von der Oelsnitz, 2025)

- Structured literature review to identify skills and traits for successful leadership in times of AI
- Identified 24 leadership competencies
- **What do you think: What will leaders need?**

Psycho-social skills:

- Awareness (**understanding oneself, others, and the situation**)
- Team member evaluation & management
- **Communication**
- Collaboration
- Empathy
- Motivation
- Networking

Psychological traits

- Adaptiveness (ability to adapt to circumstances in the environment)
- Agility (ability to understand and think quickly) and flexibility
- Solution orientation
- Being a visionary

Human resource development

- Empowerment
- Engagement (strong personal interest)
- Execution (get things done!)

Humanistic skills

- Creativity

Technical skills

- Understanding AI

AI and Business model innovation

(Costa-Climent et al., 2024; Hamilton 2025; Hayward 2025;)

- Business model = content, structure, and governance of transactions designed to create value through business opportunities
- Four central sources for value creation in the digital realm:
 - **Efficiency** (e.g., reduction of costs, improved service quality, better coordination, more productivity, more efficient delivery)
 - **Lock-in** (e.g., development of algorithms, democratisation of AI, scalable eco-system integrations)
 - **Novelty** (e.g., personalised products and services, new approaches to business models, outcome-based services)
 - **Complementary** (e.g., real-time data analytics for personalisation, ethical AI)
- But: So far, the actual win through AI is modest, adoption fails and additional work is created through AI-generated errors
- SO – will agents save us or will the AI-bubble burst?

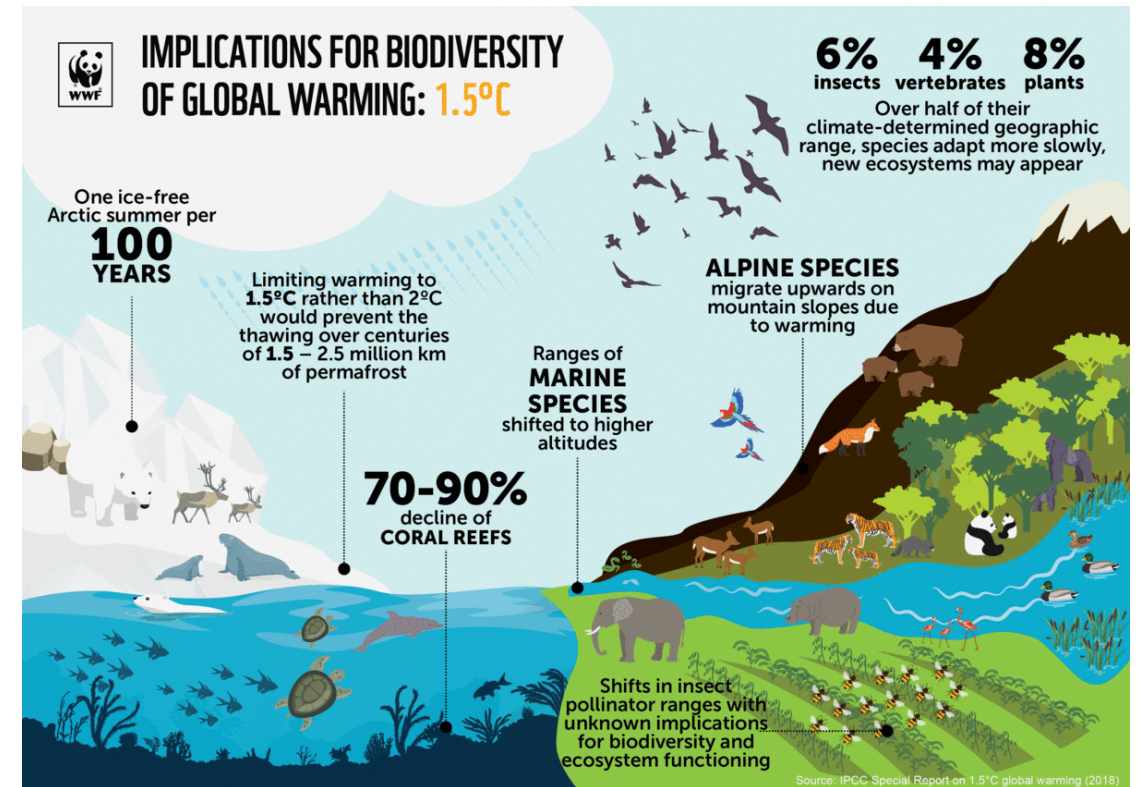
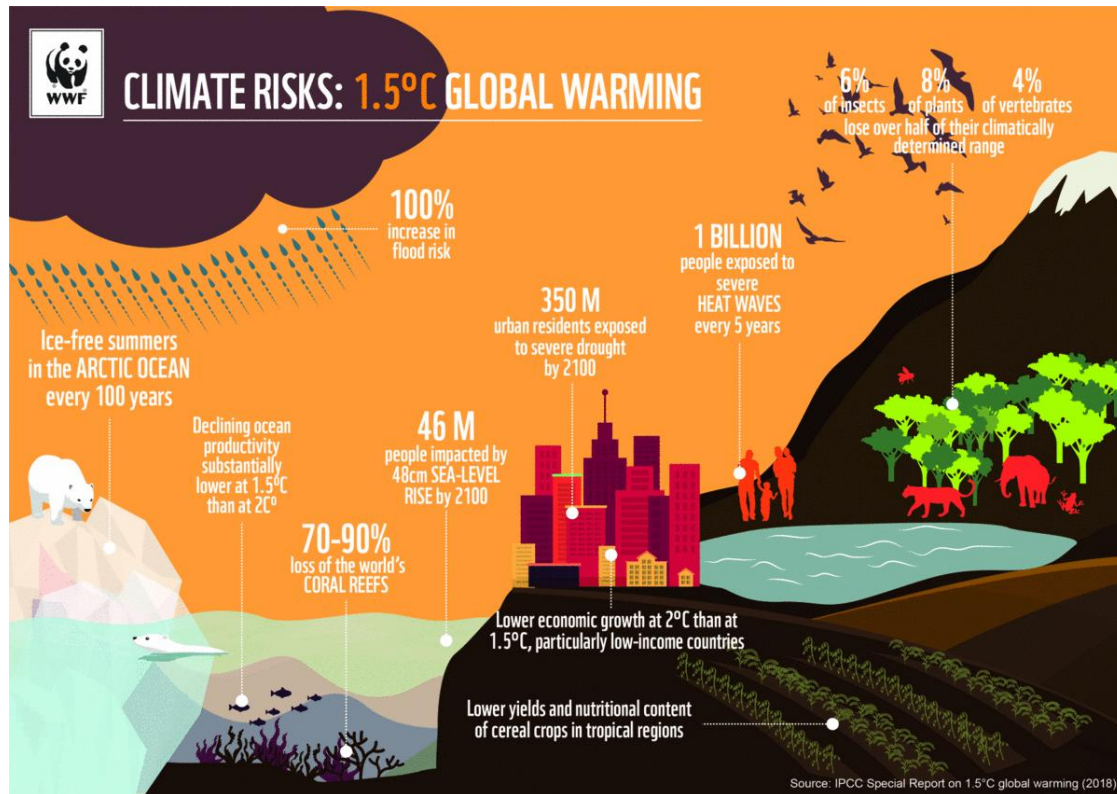


What do you think?



**One area of innovation:
AI and sustainability**

Clear is: We need to fight climate change – and that means reducing CO2 and resource preservation



https://wwf.panda.org/discover/our_focus/climate_and_energy_practice/ipcc152/

Customers care about climate change

(McKinsey & Company 2023)

- Customers care for climate friendliness: 78% of US-Americans say that a sustainable lifestyle matters to them
- The market for products that claim sustainability is growing → particularly, when it comes to food and personal care (less for beer)



Sustainability Transformation is the Next Frontier

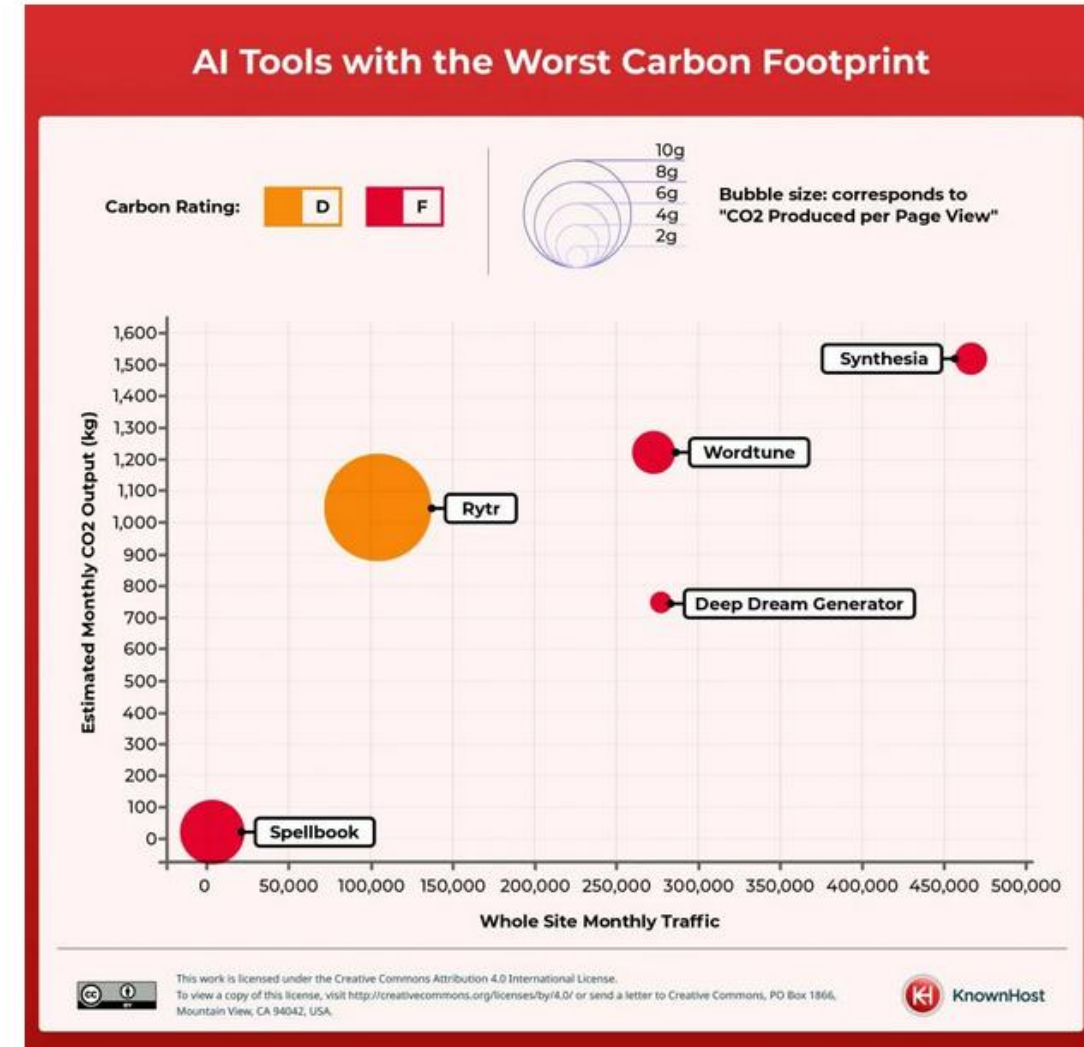
The path to a more sustainable world and a future-proofed business model runs through Sustainability Transformation practices. In this article, learn how to take the first step in the right direction.

AI does increase emissions dramatically

(Smartly, AI, n.d., KnownHost, 2020; Crimmins (2025))

- ChatGPT alone produces over 260,930 kilograms of CO₂ – around 1.5 – 4,32 grams per view
- Equivalent to the emission of 260 flights from London to New York
- That is not the worst – Rytr emits 10 times as much
- Still: 15 ChatGPT queries = 1 hour video streaming

AI Tools with the Worst Carbon Footprint



Data center exploitation



RESEARCH TOPICS ▼ PUBLICATIONS OUR METHODS SHORT READS

Home > Research Topics > Internet & Technology > Emerging Technology >...

SHORT READS | APRIL 13, 2026

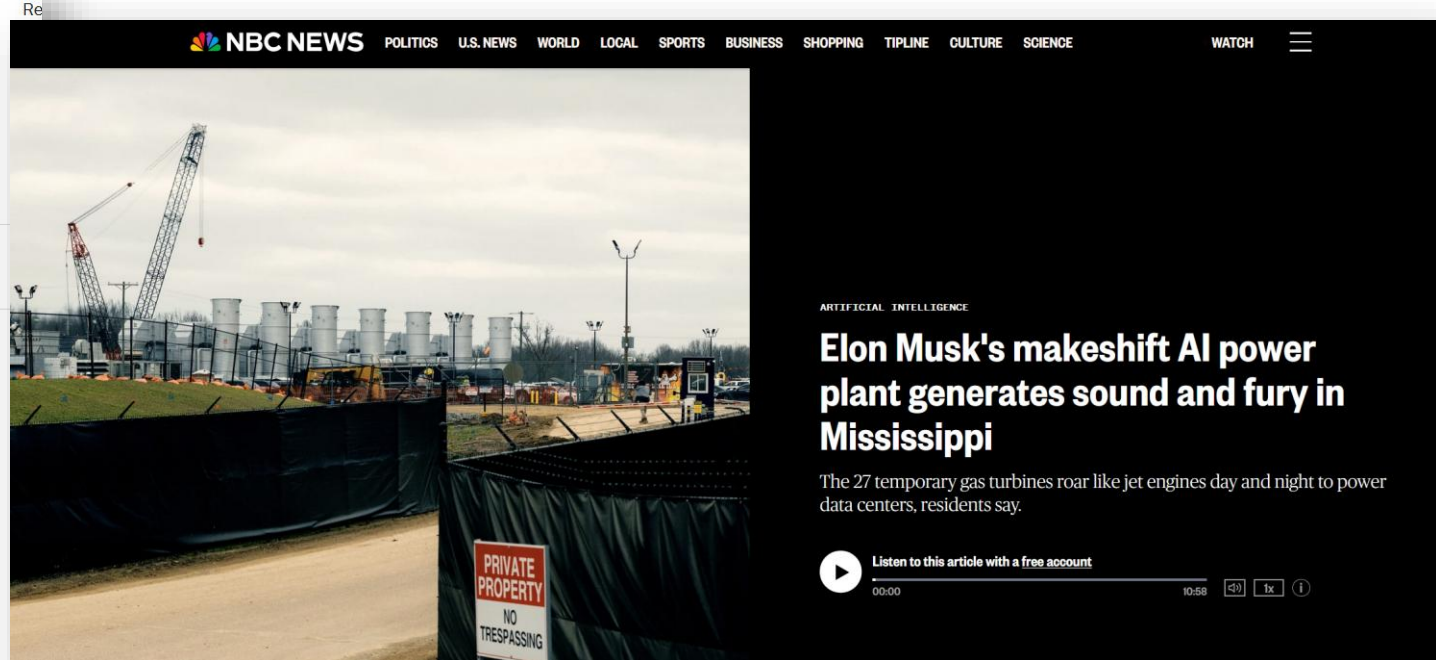
Most new data centers in the U.S. are coming to rural areas



BY SKYLER SEETS AND KAITLYN RADDE



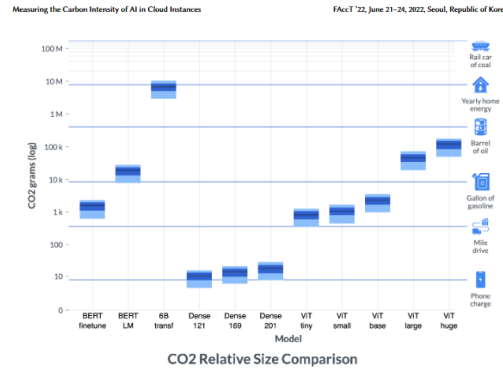
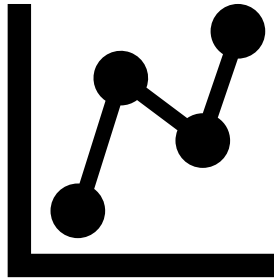
Construction continues at a new Meta data center in Beaver Dam, Wisconsin, on March 31. (Joe Timmerman/Wisconsin Watch via Getty Images)



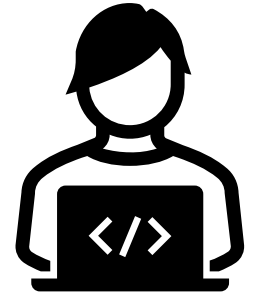
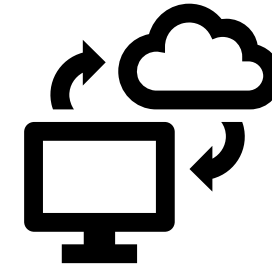
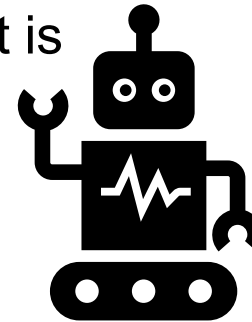
Why?

(Patterson 2022)

- How much data is the model trained on?



- How efficient is the model?



- Which computer/ cloud is used to run the model?

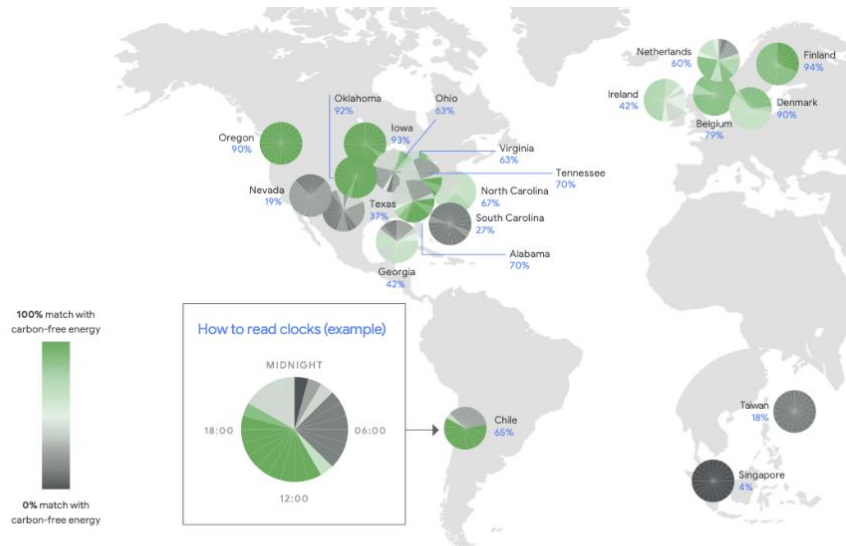
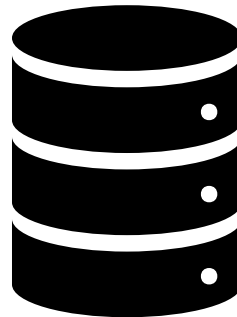


Figure 2. [Percent Carbon Free Energy by Google Cloud Location in 2020](#). The map shows the %CFE and how the percentage changes by time of day. Chile has a high %CFE from 6AM to 8PM, but not at night. The US examples on this map range from 19% CFE in Nevada to 93% in Iowa, which has strong prevailing winds both night and day. ([sustainability.google/progress/energy/](#))



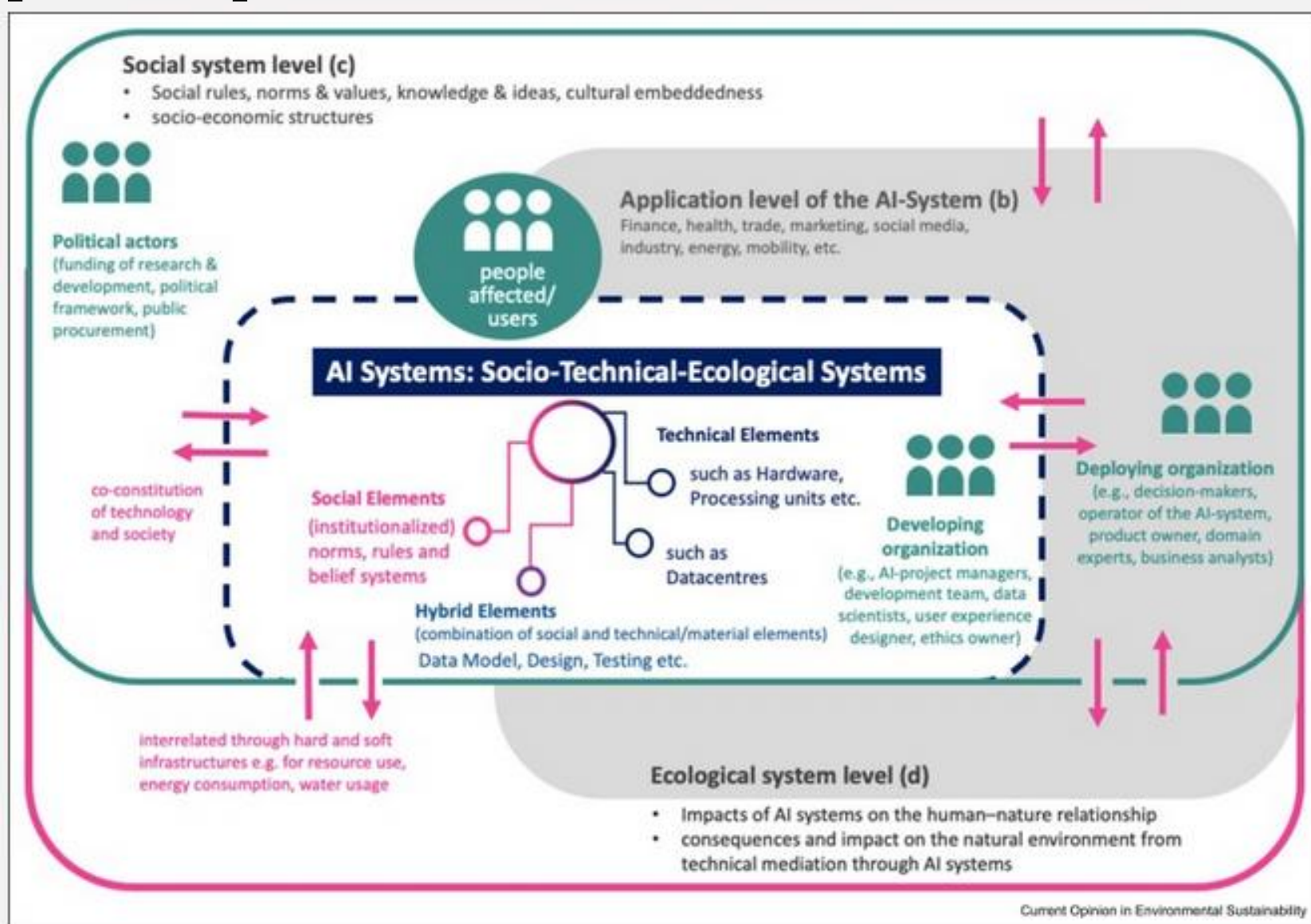
- Central: How are the data centres powered?
- Are they using sustainable energy?
- Do they need a lot of water?

Ai ***can*** be more sustainable (Patterson et al., 2022)

1. **Model:** Developing and employing energy efficient models – reduces the computation by factors of 5 to 10
2. **Machine:** Using processors optimized for ML training such as TPUs or recent GPUs (e.g., V100 or A100), versus general-purpose processors, can improve performance/Watt by factors of 2–5
3. **Mechanization.** Computing in the Cloud rather than on premise improves datacenter energy efficiency, reducing energy costs by a factor of 1.4–2
4. **Map.** Moreover, Cloud computing lets ML practitioners pick the location with the cleanest energy⁴, further reducing the gross carbon footprint by factors of 5–10

Need for a socio-techno-ecological perspective

(Rohde et al., 2024)



What are your questions?



AI 507: Artificial Intelligence and Society



#13 AI in health and medicine

Associate Prof. Dr. Lena Frischlich

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